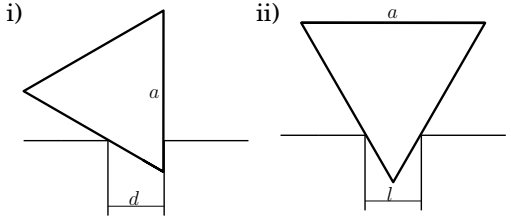
Estonian-Finnish Olympiad 2013

1. PRISM (8 points)

i) *(4 points)* A right prism that has an equilateral triangular base with length a is placed in a horizontal slit between two tables, so that one of the side faces is vertical. How small can the width d of the slit be made before the prism falls out of the slit? There is no friction between the prism and the tables and the prism is made of a homogeneous material. The edges of the slit are parallel.

ii) *(4 points)* Now the prism is placed in the slit so that one of its side faces is horizontal. How small can the width l of the slit be made before that position becomes unstable?



2. CELLPHONE CAMERA (6 points)

A photographer focussed his camera to distance L and took a photo. On the photo, all farther objects (up to infinity) turned also out to be sharp. Additionally, all closer objects down to distance s were sharp.

i) *(4 points)* What is the minimum possible L?

ii) *(2 points)* Find the corresponding s.

Background. We consider the image of a pointlike object to be sharp if its image is smaller than one pixel on the sensor. Otherwise the image is blurry. The lens of the camera may be viewed as a convex lens. The camera is focussed by changing the distance between the sensor and the lens.

Parameters. Calculate the answer for a cellphone made by a well-known company. The focal length of its camera $f = 4.3\,\text{mm}$ and the diameter of the lens $D = 1.8\,\text{mm}$. The sensor is $w = 4.6\,\text{mm}$ wide corresponding to $N = 3264$ pixels.

3. MISSION TO MARS (7 points)

A crew of astronauts is going to be sent to explore the polar region of Mars and search for buried water ice. Their spaceship will travel from Earth to Mars along an elliptic transfer orbit tangential to the orbits of both planets. Despite its shortcomings, this orbit is commonly used in space travel due to its relatively good fuel economy. Future manned missions to Mars are very likely based on this kind of transfer. In this problem you will examine some aspects of this orbit.

The mean orbital radius of Mars is $R_a = 1.52\,\text{AU}$. The mean orbital radius of Earth is $R_g = 1\,\text{AU} = 149\,600\,000\,\text{km}$. Mars has a mean radius of $r_a = 3397\,\text{km}$ and surface gravity $g_a = 3.71\,\text{m/s}^2$. Earth has a mean radius of $r_g = 6371\,\text{km}$.

i) *(1 point)* Find the orbital period T_a of Mars, i.e. find the length of Martian "year" in Earth years.

ii) *(1.5 points)* How long (t_t) does a one-way trip to Mars take?

iii) *(1.5 points)* The spaceship is put into this orbit by using a powerful rocket. It is more efficient to burn fuel as a short burst when the spaceship is still near Earth. How much additional speed (Δv_1) does the booster have to be able to give to the spaceship to enter the transfer orbit, starting from the north pole? Neglect the air resistance.

iv) *(1.5 points)* Estimate the Δv_2 needed to enter a circular orbit close to Mars.

v) *(1.5 points)* What is the minimal duration of the trip to Mars and back?

4. MAGNETIC DIPOLES (7 points)

Let us consider the following model for a magnetic dipole. Some wire with no resistance has been bent into the shape of a square with side length a. At some point on the wire is a small ideal current source that keeps current I flowing in the circuit in all situations. The magnetic moment m of a planar circuit is given by the relation $m = IA$, with vector $\vec{m}$ pointing in the normal direction of the circuit according to the right hand rule (A is the area bounded by the circuit).

i) *(3 points)* The dipole is placed inside a homogeneous magnetic field $\vec{B}$, so that the angle between $\vec{m}$ and $\vec{B}$ is θ. Find the angles θ_s and θ_u that correspond to stable and unstable equilibria, respectively. Calculate the amount of work (w) needed to rotate the dipole from θ_s to θ_u. Give your answer in terms of m and B.

We can use this model to calculate the magnetic properties of materials containing unpaired electrons that have negligibly weak interactions with one another. Let us consider a sample of material with n such unpaired electrons per unit volume, placed inside a homogeneous magnetic field $\vec{B}$. Due to spin, each unpaired electron acts as a small magnetic dipole. However, owing to the quantum nature the electron, the projection of its magnetic moment along $\vec{B}$ can only be μ_B or $-\mu_B$ (μ_B is called the Bohr magneton).

ii) *(4 points)* Calculate M, the magnetic moment per unit volume of the sample, if the temperature of the material is T and the external magnetic field is B.

5. FRICTION OF A STRING (8 points)

Measure the dynamic coefficient of friction μ_1 between the ballpoint pen and the string. Estimate the uncertainty. It might help that the dynamic coefficient of friction between the pencil and the same string was measured beforehand and $\mu_2 = 0.20 \pm 0.01$ was obtained.

Equipment: dynamometer, string, ballpoint pen, pencil and weight.

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6. SPHERE AND CYLINDER (7 points) A sphere and a cylinder are lying on an inclined surface with inclination angle α. Both have mass m and radius r. The bodies are released from equal initial heights H. The moments of inertia of the sphere and the cylinder are, respectively, $I_{\text{sph}} = \frac{2}{5}mr^2$ and $I_{\text{cyl}} = \frac{1}{2}mr^2$. The coefficient of friction between the surface and the bodies is μ.

i) *(2 points)* Which of the bodies comes down faster? What was the relative lag of the slower body $\gamma = (t_2 - t_1)/t_1$? The times t_1 and t_2, respectively, denote the traveling times of the faster or the slower body. Assume that the rolling occurs without slipping.

ii) *(2.5 points)* Find the minimal angle of inclination α_0 for which the cylinder starts to slide in addition to rolling.

iii) *(2.5 points)* If $\alpha \to 90°$, the bodies obviously lose contact with the surface and fall down in free fall with equal times. What is the minimal angle of inclination α_m, for which both the sphere and the cylinder come down with equal times?

7. BURNING WITH A LENS (7 points) Sunrays are focused with a lens of diameter $d = 10\,\text{cm}$ and focal length of $f = 7\,\text{cm}$ to a black thin plate. Behind the plate is a mirror. Angular diameter of the Sun is $\alpha = 32'$ and its intensity on the surface of the Earth is $I = 1000\,\text{W/m}^2$, Stefan-Boltzmann constant $\sigma = 5.670 \times 10^{-8}\,\text{W/(m}^2\text{K}^4)$.

i) *(4 points)* Find the temperature of the heated point of the plate.

ii) *(3 points)* Using thermodynamic arguments, estimate the maximal diameter of the lens for which this model can be used.

8. ZENER DIODE (7 points) An inductance L and a capacitor C are connected in series with a switch. Initially the switch is open and the capacitor is given a charge q_0. Now the switch is closed.

i) *(1 point)* What are the charge q on the capacitor and the current I in the circuit as functions of time? Draw the phase diagram of the system — the evolution of the system on a $I - q$ graph — and note the curve's parameters. Note the direction of the system's evolution with arrow(s).

A *Zener diode* is a non-linear circuit element that acts as a bi-directional diode: it allows the current to flow in the positive direction when a forward voltage on it exceeds a certain threshold value, but it also allows a current to flow in the opposite direction when exposed to sufficiently large negative voltage. Normally the two voltage scales are quite different, but for our purposes we will take a Zener diode with the following volt-ampere characteristics: for forward currents, the voltage on the diode is V_d, for reverse currents, the voltage on the diode is $-V_d$, for zero current the voltage on the diode is $-V_d < V < V_d$.

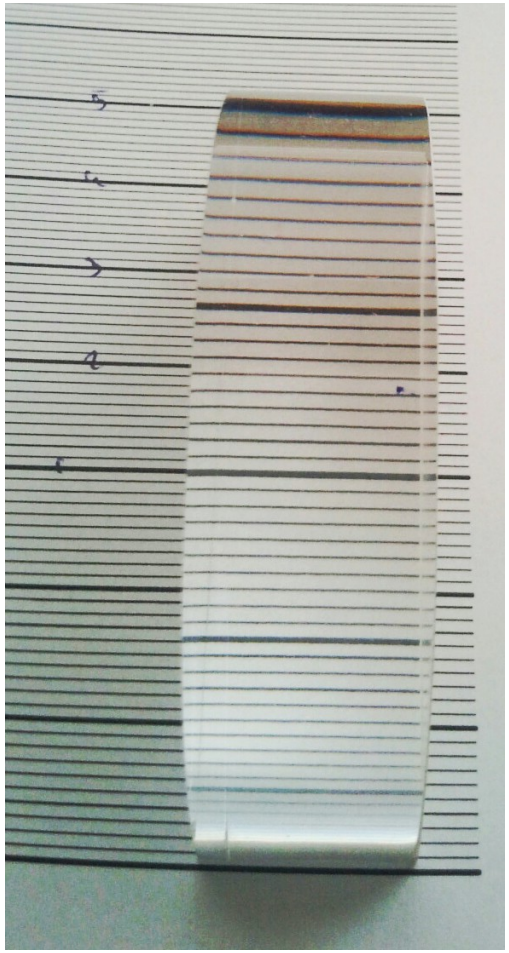
Now we connect the inductance L, the capacitor C all in series with a switch and a Zener diode. The switch is initially open. The capacitor is again given the charge $q_0 > CV_d$ and the switch is then closed.

ii) *(2 points)* Make a drawing of the phase diagram for the system. Note the direction of the system's evolution with arrow(s).

iii) *(2 points)* Does the evolution of the system only necessarily stop for $q = 0$? Find the range of values of q on the capacitor for which the evolution of the system will necessarily come to a halt.

iv) *(2 points)* Find the decrease Δq in the maximum positive value of the capacitor's charge q after one full oscillation. How long does it take before oscillation halts?

9. GLASS CYLINDER (7 points) In the following figure, there is a half-cylinder, made of glass and put on a paper with stripes (the inter-stripe distance is everywhere the same). Find the coefficient of refraction of the glass.



10. RESISTIVE HEATING (8 points) Measure the resistor. You are not asked to estimate the uncertainty.

Equipment: resistor, voltage source (batteries), ammeter, calorimeter, thermometer, stopwatch.

The calorimeter has $V = 0.80\,\text{dl}$ of water and $m_a = 27\,\text{g}$ of aluminum, specific heat capacity of water $c_w = 4.2\,\text{J/(K}\cdot\text{g)}$ and of aluminum $c_a = 0.90\,\text{J/(K}\cdot\text{g)}$. Internal resistance of the batteries will vary. Your set of batteries may become depleted, spares are available.

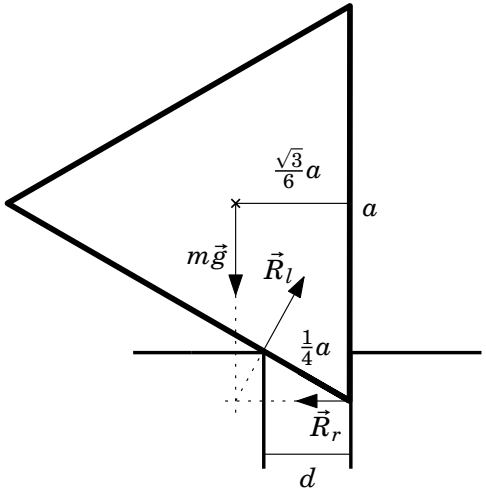
Estonian-Finnish Olympiad 2013 Solutions

1. PRISM (8 points)

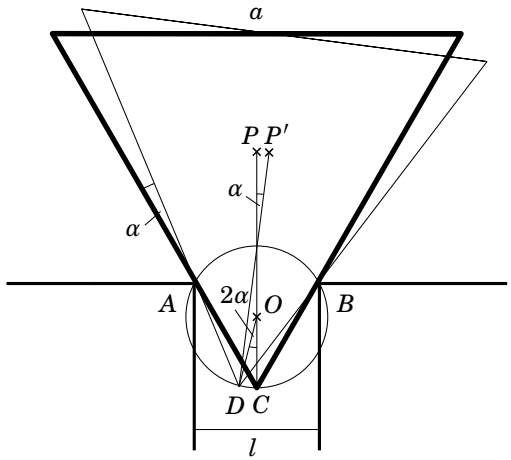
i) *(4 points)* The prism is acted on by three forces: reaction force $\vec{R}_l$ from the left-hand table, directed perpendicularly to the prism's face; reaction force $\vec{R}_r$ from the right-hand table, directed horizontally (with its point of action to be determined yet); and gravitational force mg, directed vertically and applied at the triangle's centre. (Considering a planar triangular cross-section of the prism is enough.)

On the verge of falling out, the force $\vec{R}_r$ is applied at the lower corner of the triangle. If a body in equilibrium is acted on by three forces, then their lines of action must intersect at one point. This is because otherwise the torque of one of those forces would not be zero with respect to the intersection of the lines of action of the two other forces.

As the distance between the triangle's centre and its side is $\frac{\sqrt{3}}{6}a$, the distance between the points of action of $\vec{R}_l$ and $\vec{R}_r$ is $\frac{\sqrt{3}}{6}a\cos 30° = \frac{1}{4}a$. Thus, $d = \frac{1}{4}a\cos 30° = \frac{\sqrt{3}}{8}a$.



ii) *(4 points)* Let the corners be denoted by A and B, and the tip of the prism (at its equilibrium position) by C. Let us consider a small rotation of the prism (assuming it remains in contact with the corners). The trajectory of the tip is a circle ascribed around the triangle ABC (it follows from the property of the inscribed angles because the $\angle ACB$ remains equal to 60°). The radius of that circle $r = l/\sqrt{3}$; its centre will be denoted by O. Once the prism rotates by angle α, so that the new position of the tip will be D, the central angle $\angle COD = 2\alpha$. Hence, the tip is raised by $r - r\cos(2\alpha) \approx 2r\alpha^2$. The height of the centre of mass P of the prism is raised because the tip is raised, and lowered because the vertical projection of the segment CP is reduced by $|CP|(1-\cos\alpha) \approx |CP|\alpha^2/2$. Here, $|CP| = a/\sqrt{3}$. So, the original position is stable if $\frac{a}{\sqrt{3}}\alpha^2/2 < 2\frac{l}{\sqrt{3}}\alpha^2$, hence $l > \frac{1}{4}a$.



2. CELLPHONE CAMERA (6 points)

The distance L is often called hyperfocal distance in photography and it was calculated more than one hundred years ago by Louis Derr (the figure is taken from his book *Photography for students of physics and chemistry*, published in 1906).

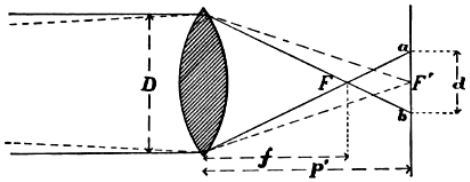


FIG. 49. Hyperfocal Distance.

Let's consider that the camera is focused to distance L and the image is formed exactly on the sensor's plane. The object's distance L and its image's distance a (corresponds to p' on the figure) are related by the lens formula $\frac{1}{L} + \frac{1}{a} = \frac{1}{f}$, thus

$$a = \frac{Lf}{L-f} = \frac{Lf}{L(1-f/L)} \approx \frac{Lf}{L}\left(1+\frac{f}{L}\right) = f + \frac{f^2}{L},$$

where the approximation $(1+x)^{-1} \approx 1-x$ (for small x) was used. Image's distance exceeds the focal length by $\Delta a = a - f = f^2/L$.

i) *(4 points)* The light coming from an infinitely far away object will pass the focal point F and form a cone which is cut by the sensor's plane. The diameter d of the cut on the sensor's plane can be found from similar triangles $d/D = \Delta a/f$, thus $d = Df/L$. Taking into account the sharpness condition $d \le \eta$, where $\eta = w/N$ is the size of a single element of the sensor, we find that the limiting value of L is $L = Df/\eta = DfN/w \approx 5.5\,\mathrm{m}$.

ii) *(2 points)* We'll now find the shortest distance s satisfying the sharpness condition. Object at distance s will have an image at distance $b = f + f^2/s$ and the light passing the lens will converge behind the sensor's plane forming a cone. The diameter d_2 of the cone's cut with the sensor's plane can be calculated from similar triangles: $d_2/D = (b-a)/b$. Accounting for sharpness condition $d_2 = \eta$, we can express $b = a/(1-\eta/D)$, and substituting the values of a and b gives

$$f + \frac{f^2}{s} = \frac{f + \frac{f^2}{L}}{1-\eta/D} = f\frac{1+\eta/D}{1-\eta/D} \approx f\left(1+\frac{\eta}{D}\right)^2 \approx f\left(1+\frac{2\eta}{D}\right).$$

Finally, $f^2/s = 2f\eta/D$, or $s = \frac{1}{2}Df/\eta = L/2 \approx 2.75\,\mathrm{m}$.

3. MISSION TO MARS (7 points)

i) *(1 point)* We can find the orbital period of Mars from Kepler's third law ${R_a}^3/{R_g}^3 = {T_a}^2/{T_g}^2$, giving $t_t \approx 1.87\,\mathrm{yr}$.

ii) *(1.5 points)* Again, we can use Kepler's third law to calculate half of the orbital period.

$$t_t = \frac{T_g}{2}\frac{(R_a+R_g)^{3/2}}{(2R_g)^{3/2}} \approx 0.707\,\mathrm{yr}.$$

iii) *(1.5 points) Background.* Δv is important, because the sum of all Δv determines how much fuel is needed for a given mission. The fuel needed is exponential of total Δv and is described by Tsiolkovsky rocket equation.

Kinetic energy per unit mass of such a transfer orbit where it intersects the Earth's orbit is $-\frac{GM_s}{R_g+R_a} + \frac{GM_s}{R_g}$. Using the orbital angular speed of Earth we can substitute $GM_s = \frac{4\pi^2 R_g^3}{T_g}$. The speed at the beginning of the transfer orbit becomes

$$v_{t_0} = \sqrt{2GM_s\left(\frac{1}{R_g} - \frac{1}{R_g+R_a}\right)} \approx 32.7\,\mathrm{km/s}$$

. The speed in Earth's inertial frame is $v'_{t_0} = v_{t_0} - v_g \approx 2.94\,\mathrm{km/h}$. To achive that, we first need to escape Earth's gravity, so

$$\Delta v_1 = \sqrt{2\left(\frac{{v'_{t_0}}^2}{2} + \frac{GM_g}{r_g}\right)}.$$

Using the surface gravity of Earth we can substitute $\frac{GM_g}{r_g} = g_g r_g$ so $\Delta v_1 \approx 11.2\,\mathrm{km/s}$.

iv) *(1.5 points)* We can calculate the speed of the transfer orbit where it intersects the orbit of Mars from Kepler's second law $v_{t_1} = v_{t_0}/1.52 \approx 21.5\,\mathrm{km/s}$. The speed of the spacecraft relative to Mars is $v'_{t_1} \approx 3.25\,\mathrm{km/s}$. The speed of the spacecraft once near Mars surface is

$$v_{t_m} = \sqrt{2\left(\frac{v'^{\,2}_{t_1}}{2} + r_a g_a\right)} \approx 5.98\,\mathrm{km/s}$$

Since the speed of low Mars orbit is $v_{e_a} = \sqrt{r_a g_a} \approx 3.55\,\mathrm{km/s}$, we need to brake for $\Delta v_2 \approx 2.43\,\mathrm{km/s}$.

v) *(1.5 points)* The Earth–Sun–Mars angle α at the launch of the mission needs to be $\alpha = \pi - w_a t_t \approx 0.77$ for the spacecraft to reach Mars. Likewise for the return trip $\beta = \pi - w_g t_t \approx -1.301$. If we go to the corotating frame of reference with earth, we can see that the minimal time between those two angles is $\frac{2\pi-\alpha+\beta}{w_a - w_g} \approx 1.96\,\mathrm{yr}$. The minimal duration of the trip is therefore longer by t_t, giving $2.67\,\mathrm{yr}$.

4. MAGNETIC DIPOLES (7 points)

i) *(3 points)* There is no torque on the square if $\theta = 0$ or $\theta = \pi$, so one of them is stable and the other unstable. If we start from $\theta = 0$ and turn the square to some θ, but keep two sides of the square perpendicular to $\vec{B}$, Lorentz forces on these two sides give a torque $\tau = -BIa \cdot a \sin\theta = -Bm\sin\theta$ towards decreasing θ. By symmetry, we get the same result if we keep the other keep two sides of the square perpendicular to $\vec{B}$. It is possible to conclude that the torque depends only on θ (at least near $\theta = 0$), not on the exact orientation of the square. Since torque acts to restore $\theta = 0$, we find that $\theta_s = 0$ and $\theta_u = \pi$. To find the work to get from θ_s to θ_u, we can again keep two sides perpendicular to $\vec{B}$ - the answer cannot depend on the path, so we choose the simplest one. Integrating $\tau = Bm\sin\theta$ from $\theta = 0$ to π gives us $w = 2Bm$.

ii) *(4 points)* Let us denote the number of electrons (per unit volume) with magnetic moment projection $+\mu_B$ as n_+ and the ones with $-\mu_B$ as n_-. Their sum is always the same, $n_+ + n_- = n$. Also, in thermal equilibrium, their ratio is given by $\frac{n_-}{n_+} = \exp\left(-\frac{2\mu_B B}{k_B T}\right)$, where k_B is Boltzmann's constant. Solving the equations, we can find n_+ and n_-. The total magnetic moment per unit volume (in the direction of $\vec{B}$) is given by $M = \mu_B(n_+ - n_-)$. After substituting,

$$M = \mu_B N \frac{1 - \exp\left(\frac{2\mu_B B}{k_B T}\right)}{1 + \exp\left(\frac{2\mu_B B}{k_B T}\right)} = \mu_B n \tanh\left(\frac{\mu_B B}{k_B T}\right).$$

Additional comments. We see B and M always have the same sign, therefore $\vec{M}$ is parallel with $\vec{B}$. This makes sense, as we saw that $\theta = 0$ ($\vec{m}$ parallel to $\vec{B}$) orientation had lowest energy. The graph of M vs B goes to $\mu_B n$ for very large B or to $-\mu_B n$ for very small B (all spins aligned with $\vec{B}$). At $B = 0$, $M = 0$ as well, since both spin orientations have the same energy. Around zero, the curve is linear, as $\tanh x \approx x$ for small x gives us $M \approx \frac{\mu_B^2 n B}{k_B T}$.

5. FRICTION OF A STRING (8 points)

Let's first calculate the difference of tension force T between two ends of a sliding string arced over a cylinder by an angle α. Furthermore, let's look at a short piece of the arc that subtends an angle $d\alpha$. On one hand, $dT = \mu\, dR$ is the friction force acting on the piece, where dR is the reaction force. On the other hand, $dR \approx T\, d\alpha$, because both ends of the piece are pulled by a force with a radial component of $T\frac{d\alpha}{2}$ (where α is measured in radians). Therefore, we get a differential equation: $dT = \mu T\, d\alpha$ or $d\ln T = \mu\, d\alpha$, whence $T = T_0 e^{\mu\alpha}$.

As a solution to the problem, we can measure the change of the tension force for different angles α (for example, $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$, 2π, $\frac{5\pi}{2}$ etc. for several turns; however, keeping the strings vertical offers better precision) and plot $\ln T$ with respect to α. The slope of the graph is the μ to be measured.

Extra solution (not as exact). Those who cannot derive the necessary formula can still do the experiment by doing the same measurements and noting from the plot that the relationship between α and T looks exponential. Thus, we can make an ansatz that $T = T_0 X^{\mu\alpha}$: as $\alpha = 0$ must correspond to $T = T_0$, we cannot reasonably write the μ anywhere else without over-complicating the formula. Now, we can re-measure the given pencil (it may be reasonably enough approximated with a cylinder here; more exact approaches exist) and conclude that $X \approx 2.7$. From there on, the calculation is the same.

6. SPHERE AND CYLINDER (7 points)

i) *(2 points)* Since no energy is lost due to friction on sliding, the change in potential energy $\Delta E_p = mgH$ is transformed to kinetic energy consisting of both translational and rotational motion. Taking into account the rolling condition $v = \omega r$, we have

$$\Delta E_p = E_k = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}kmv^2 = \frac{1}{2}(1+k)mv^2,$$

where general expression $I = kmr^2$ for moment of inertia is used. Therefore, $v^2 = 2gH/(1+k)$.

On the other hand, the bodies travel distance $x = H/\sin\alpha$ along the slope with a constant acceleration $a_\parallel$ directed parallel to the surface. Let's express $a_\parallel$ from the equation $v^2 = 2a_\parallel x$:

$$a_\parallel = g\sin\alpha/(1+k).$$

The times are now easy to calculate as $t = v/a_\parallel$, giving

$$t = \sqrt{1+k}\left(\frac{2H}{g\sin^2\alpha}\right)^{\frac{1}{2}}.$$

Replacing $k_s = \frac{2}{5}$ for sphere and $k_c = \frac{1}{2}$ for cylinder, we find that the sphere is faster by a relative factor

$$\gamma = \sqrt{\frac{1+k_c}{1+k_s}} - 1 = \sqrt{\frac{15}{14}} - 1 \approx 0.035.$$

ii) *(2.5 points)* As found in previous subpart, the acceleration's parallel component to the slope $a_\parallel$ is smaller than the contribution by gravity $g\sin\alpha$. The difference is contributed by the friction force $F_f = mg\sin\alpha - ma_\parallel$. Sliding starts, if the necessary friction reaches the maximal value $F_{\max} = \mu N = \mu mg\cos\alpha$. Equating the two expressions gives

$$mg\sin\alpha - mg\sin\alpha/(1+k) = \mu mg\cos\alpha,$$

$$\tan\alpha = \mu\frac{1+k}{k}.$$

For the cylinder the limiting angle is $\alpha_0 = \arctan(3\mu)$.

iii) *(2.5 points)* When the maximal friction force is reached, the motion goes into rolling and sliding mode, where the total force component along the surface is given by the difference of gravity and friction:

$$F_\parallel = mg\sin\alpha - F_{\max} = mg\sin\alpha - \mu mg\cos\alpha.$$

We note that the acceleration in this mode does not depend on the moment of inertia any more.

Calculating the limiting angle of slipping mode also for the sphere $\alpha_{\mathrm{sph}} = \arctan(\frac{7}{2}\mu) >$

α_0 shows that for all angles larger than $\alpha_m = \alpha_{\rm sph}$ both bodies are in the slipping mode and thus have equal accelerations and arrival times.

7. BURNING WITH A LENS (7 points) The solar energy flux which is focused by the lens to the image of the Sun can be calculated as $P = \frac{\pi}{4} d^2 I$; the image of the Sun radiates according to the Stefan-Boltzmann law with the total power $P = \frac{\pi}{4}(\alpha f)^2 \sigma T^4$. From the heat balance we obtain $\frac{\pi}{4} d^2 I = \frac{\pi}{4}(\alpha f)^2 \sigma T^4$, hence

$$T = \sqrt{\frac{d}{\alpha f}\sqrt{\frac{I}{\sigma}}} \approx 4500\,\mathrm{K}.$$

Due to the second law of thermodynamics, it is impossible to direct heat energy from a lower temperature body to a higher temperature body. Hence, the image temperature cannot exceed the temperature of the Sun. Now we can use the known temperature of Sun $T_0 = 6000\,\mathrm{K}$, but it is better to use the Stefan-Boltzmann law for solar radiation flux density: near the Sun's surface, $I_0 = \sigma T_0^4$, with the total flux of $P_t = 4\pi R_s^2 I_0$. Near the Earth, the total flux $P_t = 4\pi L^2 I$; here, R_s is the Sun's radius, and L — the orbital radius of the Earth. From here we obtain $I = I_0 R_s^2/L^2 = \sigma T_0^4 R_s^2/L^2$; using the previous result,

$$T = T_0\sqrt{\frac{d}{\alpha f}\frac{R_s}{L}}.$$

Let us note that $\alpha L = 2R_s$, hence

$$T = T_0\sqrt{\frac{d}{2f}} \le T_0,$$

which means that $d \le 2f$.

8. ZENER DIODE (7 points)

i) *(1 point)* Kirchoff's 2nd law gives $L\dot{I} + q/C = 0$ or $\ddot{q} + \frac{1}{LC}q = 0$. This is the equation of a simple harmonic oscillator with the frequency $\omega = \frac{1}{\sqrt{LC}}$ and we can immediately write $q(t) = q_0 \cos\omega t$, while $I(t) = \dot{q}(t) = -\omega q_0 \sin\omega t$.

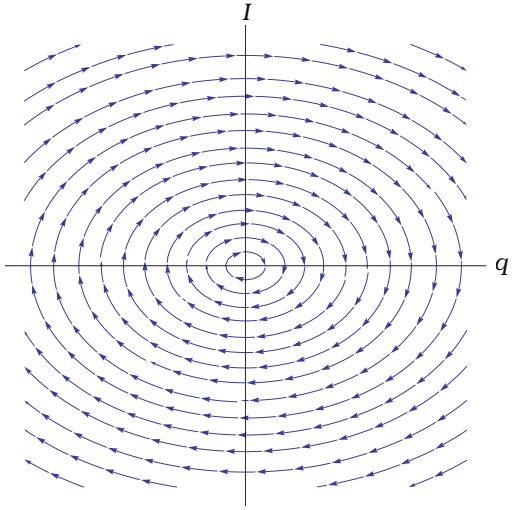
Note that

$$q^2 + \frac{1}{\omega^2} I^2 = q_0^2(\sin^2\omega t + \cos^2\omega t) = q_0^2,$$

and therefore the phase diagram of the system is an ellipse centred at the origin, with semi-axes q_0 and ωq_0. Alternatively, this relation comes directly from the conservation of energy:

$$\frac{LI^2}{2} + \frac{q^2}{2C} = E_0 = \frac{q_0^2}{2C}.$$

By looking at q and I a quarter-period later from $t = 0$, say, it's not hard to see that the system must evolve in a clockwise sense on the phase diagram. Note that in this instance, only $q = 0$ is an equilibrium point: for all non-zero q there will be never-ending oscillations in the circuit.



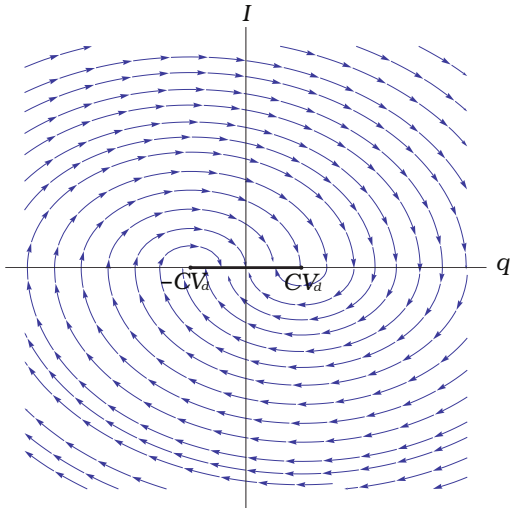
ii) *(2 points)* Now the sign of the voltage on the diode depends on the direction of the current, giving either of $L\ddot{q} + \frac{q}{C} \pm V_d = 0$. We can summarize the equations as follows:

$$L\ddot{q} + \frac{q}{C} = V_d \qquad \text{if } \dot{q} < 0$$
$$L\ddot{q} + \frac{q}{C} = -V_d \qquad \text{if } \dot{q} > 0$$

Let us introduce the new variables $q_{1,2}$ such that $q_1 = q - CV_d$ and $q_2 = q + CV_d$. Then we can rewrite the two equations above in a more familiar form:

$$L\ddot{q}_1 + \frac{q_1}{C} = 0 \qquad \text{if } \dot{q} < 0$$
$$L\ddot{q}_2 + \frac{q_2}{C} = 0 \qquad \text{if } \dot{q} > 0$$

Thus the introduction of the diode only serves to shift the equilibrium points for the otherwise simple harmonic orbits. For $\dot{q} > 0$, the equilibrium point is $q_2 = 0$ or $q = -CV_d$, while for $\dot{q} < 0$ it is $q = CV_d$. So the orbit will consist of half-ellipses in the upper and the lower parts of the $I - q$ diagram, centred at $q = -CV_d$ for the upper half and at $q = CV_d$ for the lower half. As the evolution is continuous, these half-ellipses will join up at $I = 0$.



iii) *(2 points)* We can see on the diagram that there is a "dead zone" between $\pm CV_d$ (for $I = 0$). If a trajectory reaches any of the points in that segment, it will stay there forever. The extent of that region is $2CV_d$.

iv) *(2 points)* Let's use the phase diagram to figure this out. Suppose the capacitor initially has the charge $q_0 \gg CV_d$. Then the charge will first swing to the other way of CV_d and will become $q_{T/2} = CV_d - (q_0 - CV_d) = 2CV_d - q_0$. Then it will perform the other half-oscillation around $-CV_d$ and the charge at the end of that is $q_T = -CV_d + (-CV_d - (2CV_d - q_0)) = q_0 - 4CV_d$, and therefore $\Delta q = -4CV_d$.

Note that we have the right to talk about half- and full periods because the oscillations still happen at the immutable frequency $\omega = \frac{1}{\sqrt{LC}}$. Therefore the time between the two maxima is just a full period of oscillation, $T = \frac{2\pi}{\omega}$.

Once $q(t)$ has a zero derivative inside the region bounded by $\pm CV_d$, it will remain at that particular value forever. For a large initial q_0, we expect there to be approximately $\left|\frac{q_0}{\Delta q}\right| = \frac{|q_0|}{4CV_d}$ total oscillations.

More exactly, the distance from the "dead zone" is initially $|q_0| - CV_d$ and decreases during each half-oscillation by $2CV_d$. The total number of half-oscillations is $N = \left\lfloor \frac{|q_0| - CV_d}{2CV_d} \right\rfloor$ and the total time $t = N\frac{T}{2} = N\frac{\pi}{\omega} = N\pi\sqrt{LC}$.

9. GLASS CYLINDER (7 points) The axis of the half-cylinder is where the stripe and its image coincide (form a straight line). The front edge of the half-cylinder is at the 28th line, counting from the axis, hence the radius of the cylinder $R \approx 28$. Let us consider the refracting ray s which is very close to a total internal reflection. One can see the images of 20 lines (ca 20.2, to be more precise), when counting from the central line upwards; the upper edge of the half-cylinder coincides with

the 42nd line at the background.

So, the ray s arrives at the camera at the angle $\alpha = \arcsin(28/42) \approx 41.8^\circ$ with respect to the plane of the paper. The projection of the refraction point to the paper surface lies at the distance $a = 28\sin\alpha = 28^2/42 \approx 18.7$ lines from the axis. Therefore, before refraction, ray s forms an angle $\beta = \arcsin[(20.2-18.7)/(28\cos\alpha)] \approx 4.1^\circ$ with the perpendicular of the paper surface. The incidence angle of the ray s is $\alpha+\beta \approx 45.9^\circ$, hence $n = 1/\sin(\alpha+\beta) \approx 1.39$.

10. RESISTIVE HEATING (8 points) After noting the temperature of the calorimeter, connect the batteries, resistor (in the calorimeter) and ammeter in series. Choose a convenient time interval Δt and note the ammeter reading until batteries are depleted. Measure the temperature at the end, after waiting a bit or stirring the calorimeter. We want to get maximum temperature difference for precise measurement. Since $P = RI^2$,

$$\frac{Q}{R} \approx \sum_n \Delta t (I_n/2 + I_{n+1}/2)^2,$$

$$R = \frac{(c_a m_a + c_w m_w)\Delta T}{\sum_n \Delta t (I_n/2 + I_{n+1}/2)^2}.$$

The resistance used was $R = 0.47\,\Omega \pm 5\%$. In the described circuit the batteries were depleted in 10 to 15 minutes and the temperature of the calorimeter rose by 7 to 10 degrees.